IST 691: Research Methodology and Scientific Writing Skills)

Problem Identification and Hypothesis Formation

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Outline

- Sources of Research Ideas
- Review of Literature
- Formulating the Research Problem
- Fomulating Hypotheses

(Some) Sources of Research Ideas

- Everyday life
- Practical issues
- Past research
- Theory

Everyday Life

- Our everyday experiences can spark the need for research in an area
  - What can be done to improve the efficiency/effectiveness of a piece of software/a user interface/piece of equipment, etc.?
  - How can available data be harnessed to improve on management and decision making?

Practical Issues

- Solving problems faced by private businesses, the hospitals, etc.

Past Research

- Research aims to provide answers to identified problems
- But research may also (and usually) asks more questions, based on findings

Theory

- Theory
  - Group of logically organized (i.e., deductively related) laws
  - Is both a tool and a goal
  - As a goal
    - Theories summarize and integrate existing knowledge
  - As a tool
    - Theories guide research
    - A good theory goes beyond the goal function to suggest new relationships and make new predictions
    - Thus, serves as a source of researchable ideas

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Review of Literature

- Before and after obtaining a research topic, you need to do a literature review
- Reviewing the literature early in the research process can help in several ways
  - It can tell you if a research question has already been answered
  - It can help you evaluate the interestingness of a research question
  - It can give you ideas on how to conduct your own study
  - It can tell you how your study fits into the research literature

Review of Literature

- The research literature in any field is all the published research in that field
- In general, research literature does not include self-help books, dictionary and encyclopedia entries, websites, and similar sources that are intended mainly for the general public
  - These are considered unreliable because they are not reviewed by other researchers and are often based on little more than common sense or personal experience
  - Wikipedia for example, contains much valuable information, but the fact that its authors are anonymous and may not have any formal training or expertise in that subject area, and its content continually changes makes it unsuitable as a basis of sound scientific research
- Good sources of research literature
  - Academic journal articles, books, conference proceedings, association papers, dissertations, etc., written by scholars and researchers for scholars and researchers

Starting the Literature Review

- Be familiar with the library
- Be prepared to spend considerable amounts of time in the search for literature
- Journal article abstract
  - Gives you information about the content of an article so you can tell whether to select the article for further reading
- For journal article you select to read further
  - Take detailed notes on its contents
    - Complete reference, methodology, important findings, strengths, weaknesses, other thoughts and comments
    - Electronic citation and reference tools exist to help you keep track of your references

Defining Objectives

- What are you doing the literature search for?
  - To familiarize you with the topic you want to investigate?
  - To help develop a methodology you need to use in conducting your research?
- Identifying your objectives helps you focus the literature search

Doing the Search

- Several sources for material
  - Books
  - Journals and conference proceedings
  - Electronic databases
  - Internet resources

Doing the Search

- Books
  - Good place to start literature search
  - Provide you with an introduction to your research topic and a summary of published literature up to the time of writing the book
- Computerized and electronic databases
  - UB Library has access to some of these
- Internet resources
  - Email, discussion groups, web searches (especially of academic sites)

Doing the Search

- Journal articles and conference proceedings
- Key difference between a journal and a conference article
  - Journal article is a longer piece of writing which has a definite structure and is published in journals
  - Conference paper is generally a short and a more precise paper that is presented at a conference

Doing the Search

- Most articles in professional journals are one of two basic types
  - Empirical research reports
    - Describe one or more new empirical studies conducted by the authors
    - They introduce a research question, explain why it is interesting, review previous research, describe their method and results, and draw their conclusions
  - Review articles
    - Summarize previously published research on a topic and usually present new ways to organize or explain the results
    - When a review article is devoted primarily to presenting a new theory, it is often referred to as a theoretical article

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Formulating the Research Problem

- Research problem
  - Researcher must make clear the research problem
    - An exact statement of the specific problem to be investigated
  - Research problem must be identified before any data are collected

Defining the Research Problem

- Research problem
  - A specific issue, gap in knowledge, or area of concern that a study aims to investigate, solve, or explain, forming the foundation for research questions and objectives
- Research question
  - The how
    - i.e., the specific, focused inquiry derived from the research problem, guiding exactly what you will study to address it
  - An interrogative sentence or statement that asks *what relationship exists between two or more variables?*

Defining the Research Problem

- 3 criteria for good research question
  - There should be a relationship between the variables
    - Clearly express a link, association, or cause-and-effect relationship between two or more variables being studied
  - The problem should be stated in question form (or as a clear, unambiguous statement)
    - Problem should be presented as an interrogative sentence or a clear, specific, declarative statement (in the case of a hypothesis) to minimize misinterpretation
  - The problem statement should be such as to imply possibilities of empirical testing
    - The variables and the proposed relationship must be measurable in the real world (using data collection and analysis)
    - This distinguishes a researchable problem from a philosophical or moral question

Specificity of the Question

- Specificity of the research question
  - The specificity with which the research question is stated
- Consider the following two research questions
  - What effect does the environment have on learning ability?
    - Meets all the criteria of a problem, but is too vague
  - What effect does the amount of exposure to words have on the speed with which they are learned
    - Much more specific
- How specific to we get?
  - Specific enough for the researcher to be able to have a grasp of the variables to be investigated and (in the case of an experiment) to help the experimenter in designing and carrying out the experiment

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Hypothesis

- A specific, testable prediction about the relationship between variables
  - Serves as a tentative answer to a research question
  - Grounded in existing knowledge
  - Formulated to be supported or refuted by data from the research, often using "if-then" statements to link independent and dependent variables

Hypothesis

- Charateristics
  - Testable & Falsifiable
    - Must be able to collect data to prove or disprove it.
  - Specific & Clear
    - Identifies variables (independent and dependent) and the expected relationship.
  - Educated Guess
    - Based on prior theories and knowledge, not just a random guess.
  - Predictive
    - States the expected outcome of the study.
  - Often Directional
    - Suggests the direction of the relationship (e.g., "higher grades," "faster recovery")

Hypothesis

- Types of Hypotheses (in Statistical Testing)
- Null Hypothesis (H_0)
  - States there is no relationship or difference between variables
- Scientific (Alternative) Hypothesis (H_a)
  - States the researcher's prediction of a relationship or effect (what they expect to find)
- In an experimental study, it is the null, and not scientific hypothesis that is tested
  - The scientific hypothesis does not specify the exact amount or type of influence to be expected
  - To obtain support for the scientific hypothesis, you must collect evidence that enables you to reject the null hypothesis

Hypothesis

- Example
  - Research Question: Does studying more improve test scores?
  - Hypothesis: Students who study at least 8 hours a week will score higher on average on standardized tests than those who study less.
  - Independent Variable: Hours of study.
  - Dependent Variable: Test scores

Formulating the Hypothesis

- In an experiment, it is the hypotheses derived from the research questions that are tested, and not the research questions themselves
- Hence, in all cases
  - A hypothesis must be stated so that it is capable of being either refuted or confirmed
- A hypothesis that fails to meet the criterion of testability removes the problem from the realm of science, and any conclusion reached does not represent scientific knowledge